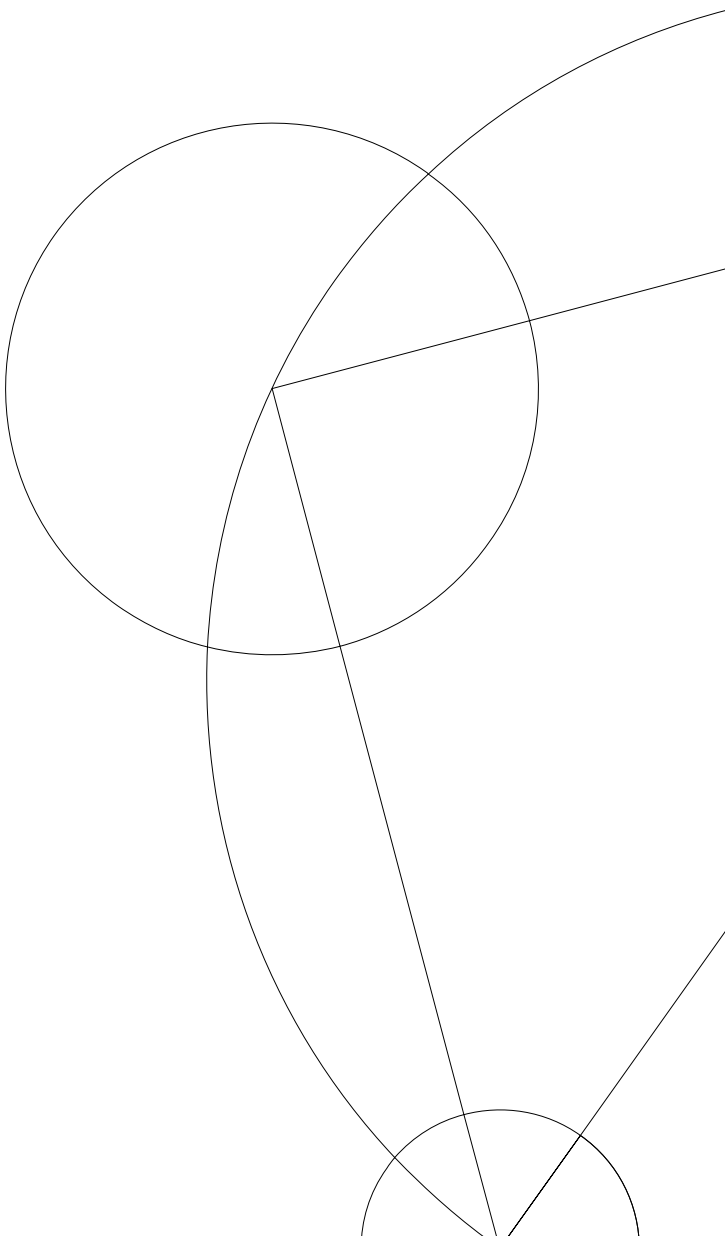
Econometrics II

Assignment 1 Group 69



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1 Introduction

US housing fell historically low between 2007 and 2009, reaching 526.000 units in 2009(1). This paper asks one question: *using only information available in the first quarter of 2009, could a univariate time series have forecasted the subsequent recovery?* We model quarterly US housing starts in logs using data from 1959(1)-2009(1).

Based on the in-sample diagnostics, an $AR(6)$ model is our preferred specifications. However, its dynamic forecast overestimates the speed of recovery. A simpler $AR(3)$ model performs better out of sample but also predicts a faster recovery than observed. This suggests that the historical adjustment of housing starts towards their long-run level did not continue in the same way after 2009.

2 Data

The data contains total new privately-owned housing units started, adjusted at an annual rate (in thousands), from Federal Reserve Bank of St. Louis, quarterly for 1959(1)-2026(2), with $T = 270$ observations. The lowest observation in the dataset is 525, and highest being 2424, with the data yielding a mean of 1430, and a standard deviation of 369. We transform the housing data to log form, and define the following:

$$y_t = \log(HOUST_t) \tag{1}$$

This stabilizes the variance of a series that moves by a factor of four. The estimation sample is 1959(1)-2009(1), where 2009(2)-2019(4) and is not used in estimation, hence the report replicates a 2009 forecaster's information. Furthermore, taking the log of a dependent variable can potentially reduce or totally remove heteroskedasticity. This will be analyzed later in section 4.3. First, we analyze the log transformation (Figure 1) and assume our process stationary, but also remain critical towards this graphical analysis, and do further testing later in Section 4.2.
Inspection of the ACF and PACF for 1959(1)-2009(1) (Figure 2 and 3) suggests an autoregressive structure. We therefore proceed with an AR specification, which is presented in Section 3.1

3 Theory

3.1 The Model

Based on the transformed series defined in Section 2.0, we consider an autoregressive model of order p

$$y_t = \delta + \theta_1 y_{t-1} + ... + \theta_p y_{t-p} + \varepsilon_t \text{ where } \varepsilon_t \sim N(0, \sigma^2) \tag{2}$$

Where y_t is observed for $t = 1959(1), \ldots, 2009(1)$, and $\delta, \theta_1, \ldots, \theta_p, \sigma^2$ are unknown. Regarding the characteristics function, $\theta(z) = 1 - \theta_1 z + ... + \theta_p z^p$ is our condition that our model is stationary. The process is stationary if the roots of the characteristics function are greater than one:

$$\mid z_j \mid > 1 \forall z_1, , , , , z_p \in \mathbb{C}$$

Where the inverse roots are

$$\mid \lambda_j \mid = \mid \frac{1}{z_j} \mid < 1$$

Stationarity in our data is extremely important, as it allows the use of law of large numbers (LLN) and the central limit theorem (CLT) to be applied. LLN ensures consistency of the estimators, while CLT ensures that they are asymptotically normal:

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \sigma^2)$$

This justifies the asymptotic distributions of the test statistics used in the analysis (Table 4)

3.2 Estimator, assumptions & inference

We assume under Gaussian error, $\varepsilon_t \xrightarrow{d} N(0, \sigma^2)$ the conditional log-likelihood of (2) is:

$$\log L(\delta, \theta_1, \ldots, \theta_p) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}(y_t - \delta - \theta y_{t-1} - \cdots - \theta_p y_{t-p})^2 \tag{3}$$

where $\varepsilon_t = y_t - \delta - \theta y_{t-1} - \cdots - \theta_p y_{t-p}$. When maximizing (3) wrt. $(\delta, \theta_1, \ldots, \theta_p)$, we are minimizing the residual sum of squares, so that the residual ML estimator solves:

$$(\hat{\delta}, \hat{\theta}_1, \ldots, \hat{\theta}_p) = \arg\min_{(\delta,\theta_1,\ldots,\theta_p)\in\Theta} \sum_{t=p+1}^{T}(y_t - \delta - \theta_1 y_{t-1} - \cdots - \theta_p y_{t-p})^2 \tag{4}$$

This coinciedes with the OLS of y_t on a constant its first p lags, with;

$$\hat{\sigma}^2 = \frac{1}{T-k}\sum_{t=0}^{T}\hat{\varepsilon}_t^2$$

[1]Because the regressors are lagged values of y_t, the estimates are biased, and consistency requires (A1) stationarity and (A2) no residual autocorrelation. If (A2) fails y_{t-j} correlates with ε_t and estimators $\hat{\theta}_j$ are inconsistent. Given (A1) and (A2), the estimates are asymptotically normal and t-statistics $t_{stat} \xrightarrow{d} N(0,1)$(Nielsen, 2025, p. 66). Homoskedasticity (A3) is needed for the standard error and forecast intervals. If it fails, the estimates stay consistent, but the reported standard error is wrong. Normality (A4) is needed for the exact finite- sample inference. Finally (A5) parameter constancy is required for the model to say anything outside the estimation sample. Table 1 collects every test used in the analysis with its null hypothesis, test statistics and asymptotic distribution under the null hypotheses.

Table 1: Misspecification tests used in the analysis. All distributions are asymptotic under H_0. R^2 refers to the auxiliary regression given in the notes, k is the number of regressors excluding the constant, and q is the number of excluded lags.

Test	Null hypothesis	Statistic	Distribution under H_0
Lag significance t-test	$\theta_p = 0$	$\xi_\theta = \hat{\theta}_p / \widehat{\mathrm{se}}(\hat{\theta}_p)$	$N(0,1)$
No autocorrelation 1–5[a]	$\gamma_1 = \cdots = \gamma_5 = 0$	$\xi_{\text{auto}} = T \cdot R^2$	$\chi^2(5)$
No ARCH 1–4[b]	$\gamma_1 = \cdots = \gamma_4 = 0$	$\xi_{\text{arch}} = T \cdot R^2$	$\chi^2(4)$
No heteroskedasticity[c]	$\psi_1 = \cdots = \psi_k = \delta_1 = \cdots = \delta_k = 0$	$\xi_{\text{het}} = T \cdot R^2$	$\chi^2(2k)$
Normality[d]	$S = 0,\ K - 3 = 0$	$\xi_{\text{JB}} = \frac{T}{6}S_T^2 + \frac{T}{24}(K_T - 3)^2$	$\chi^2(2)$
RESET23[e]	$\gamma_1 = \gamma_2 = 0$	$\xi_{\text{reset}} = T \cdot R^2$	$\chi^2(2)$

[a] Breusch–Godfrey: $\hat{\epsilon}_t = x_t'\delta + \gamma_1\hat{\epsilon}_{t-1} + \cdots + \gamma_5\hat{\epsilon}_{t-5} + u_t$.
[b] $\hat{\epsilon}_t^2 = \gamma_0 + \gamma_1\hat{\epsilon}_{t-1}^2 + \cdots + \gamma_4\hat{\epsilon}_{t-4}^2 + u_t$.
[c] $\hat{\epsilon}_t^2 = \psi_0 + \sum_{i=1}^{k} x_{it}\psi_i + \sum_{i=1}^{k} x_{it}^2\delta_i + u_t$.
[d] Jarque–Bera: $S_T = T^{-1}\sum_{t=1}^{T} e_t^3$ and $K_T = T^{-1}\sum_{t=1}^{T} e_t^4$, where $e_t = (\hat{\epsilon}_t - \bar{\epsilon})/\hat{\sigma}$ is the standardized residual.
[e] $\hat{\epsilon}_t = x_t'\delta + \gamma_1\hat{y}_t^2 + \gamma_2\hat{y}_t^3 + u_t$, where $\hat{y}_t = x_t'\hat{\beta}$.

Provided the assumptions hold, θ_j measures the effect of y_{t-j} on y_t, holding the other lags constant. As the model is specified in logs, θ_j is an elasticity: a 1% increase in new privately owned housing units in period $t-j$ is associated with a θ_j % change in period t

3.3 Forecasting

Under stationarity the unconditional mean exists, and thus converges to:

$$E[y_t] = \frac{\delta}{1 - \sum_{j=1}^{p}\theta_j} = \mu \tag{5}$$

[1]The notation $Aj, j = 1, ..., 5$ refers to assumptions.

and the y_t can be written in the $MA(\infty)$ representation:

$$y_t = \mu + \varepsilon_t + \alpha_1\varepsilon_{t-1} + \cdots + \alpha_p\varepsilon_{t-p}$$

Where the coefficients decay at the rate of the largest inverse root. Equation (5) determines the long-run forecast level, while the half life:

$$H_{\frac{1}{2}} = \frac{\log(0.5)}{\log(\lambda_{max})} \tag{6}$$

determines the convergence dynamics. Dynamic forecasts are generated recursively from our estimated model. Starting from forecast origin T, the "one-step-ahead" forecast uses observed lagged values, while the forecasts replace unavailable future values by previous forecasted values. Hence, the forecast path is generated continuously without using realized observations beyond the forecast's start. Under stationarity the forecast converges towards the unconditional mean, μ.
The forecast error variance increases with the forecast periods and converges towards the unconditional variance of y_t. 95% prediction intervals are given by;

$$\hat{y}_{t+h|T} \pm 1.96\sqrt{FEV(h)} \tag{7}$$

This assumes (A3) & (A4) hold and known parameters, so it underestimates true uncertainty

4 Empirical analysis

4.1 Lag length

Our process of GETS starts from an AR(8) since ACF is positive and falls to 8 lags (Figure 2 and 3). Using two years of lags ensures that all models are estimated on the same sample, 1961(1), making the model selection criteria comparable (Table 2) A general to specific reduction eliminates (with at 5% significance level) y_{t-8}, ($t = -1.35$) and y_{t-7},($t = 0.555$). We have y_{t-6} ($t = -2.90$ and $p = 0.0042$). Therefore, we estimate an $AR(6)$ model. AIC selects AR(6), while SC and HQ select AR(2) and AR(3), respectively (Table 2). We therefore consider these three specifications.

4.2 Is stationarity defensible?

To assess (A1), we first inspected y_t graphically (Figure 1). Although it exhibits some stationary characteristics, stationarity is not unambiguous from the plot. We therefore examine the characteristic roots of our $AR(p)$ models (Table 3). For all candidate models, the roots satisfy stationarity, supporting (A1).

Table 2: Model Selection Criteria

p	$\log L$	AIC	SC	HQ	t on y_{t-p}
1	196.35	−2.0140	−1.9802	−2.0003	**34.2**
2	209.53	−2.1402	**−2.0895**	−2.1197	**−5.27**
3	212.12	−2.1567	−2.0890	**−2.1293**	**−2.27**
4	212.35	−2.1487	−2.0642	−2.1145	0.679
5	212.75	−2.1425	−2.0410	−2.1014	−0.873
6	217.00	**−2.1762**	−2.0579	−2.1283	**−2.90**
7	217.16	−2.1675	−2.0323	−2.1127	0.555
8	218.12	−2.1670	−2.0149	−2.1054	−1.35

Table 3: Roots of the AR Lag Polynomials

Model	Root	Real	Imaginary	λ
AR(6)	1	0.89672	0.19056	0.91675
	2	0.89672	-0.19056	0.91675
	3	0.31932	0.66645	0.73900
	4	0.31932	-0.66645	0.73900
	5	-0.58822	0.36838	0.69405
	6	-0.58822	-0.36838	0.69405
AR(3)	1	0.77141	0.11103	0.77936
	2	0.77141	-0.11103	0.77936
	3	-0.28454	0.00000	0.28454
AR(2)	1	0.88799	0.00000	0.88799
	2	0.42584	0.00000	0.42584

4.3 Misspecification testing

As reported in (Table 4), the null of no residual autocorrelation is rejected for AR(2) ($p = 0.01$), hence we discard it. Meanwhile , AR(3) ($p = 0.13$) and the AR(6) ($p = 0.64$) pass; the residual autocorrelations of the latter lie within the bands (Figure 6) We retain our AR(6) model, prioritizing well-specified residuals

	AR(6)(D)	AR(6)	AR(3)	AR(2)
Constant	0.7086	0.8265	0.604	0.4674
	(2.55)	(2.81)	(2.32)	(1.8)
logHOUST_1	1.283	1.256	1.258	1.314
	(14.6)	(13.6)	(13.5)	(13.6)
logHOUST_2	−0.127	−0.1524	−0.1684	−0.3781
	(−1.05)	(−1.27)	(−1.43)	(−4.25)
logHOUST_3	−0.3441	−0.2909	−0.1728	.
	(−2.45)	(−2.13)	(−1.97)	
logHOUST_4	0.1176	0.1051	.	.
	(0.829)	(0.726)		
logHOUST_5	0.1444	0.1904	.	.
	(0.988)	(1.27)		
logHOUST_6	−0.1708	−0.2211	.	.
	(−2.15)	(−2.47)		
I:1974(3)	−0.2	.	.	.
	(−7.93)			
I:1980(3)	0.3348	.	.	.
	(5.81)			
$\hat{\sigma} = se(\hat{\varepsilon})$	0.07548	0.08007	0.08147	0.08235
Log-lik.	229.448	217.003	212.118	209.529
AIC	-2.284	-2.176	-2.157	-2.140
HQ	-2.223	-2.128	-2.129	-2.120
SC/BIC	-2.132	-2.058	-2.089	-2.089
No autocorr. 1–5	[0.13]	[0.64]	[0.13]	[0.01]
No hetero.	[0.00]	[0.00]	[0.00]	[0.00]
Normality	[0.26]	[0.00]	[0.00]	[0.01]
No Arch	[0.00]	[0.04]	[0.00]	[0.00]
RESET23	[0.10]	[0.10]	[0.15]	[0.09]
T	193	193	193	193
Sample start	1961(1)	1961(1)	1961(1)	1961(1)
Sample end	2009(1)	2009(1)	2009(1)	2009(1)

Table 4: The table shows estimates of the model from (2) with various restrictions imposed. T-ratios in () and p-values in [·] for misspecification tests.

over the parsimonious models, favored by HQ and SC. RESET23 does not reject correct functional form at the 5% level ($p = 0.10$). The null hypotheses of no heteroskedasticity ($p = 0.00$) and no ARCH ($p = 0.04$) are both rejected, hence inference is based on HCSE (Robust). Normality is rejected ($p = 0.00$), which we investigate further in Section 4.4.

4.4 Unusual observations

Non-normality appears to be driven by a few outliers, with the largest standardized residuals occurring in 1980(3) (3.94), 2008(4) (2.89) and 1974(3) (2.41) (Figure 6). These observations coincide with the lifting credit control in 1980 (Carter, 1980), the financial crisis in 2008 (Mishkin, 2011), and the oils-shock recession 1974 (Corbert, 2013). Although dummy variables for 1980(3) and 1974(3) restore residual normality (Table 4), we retain these observations as they capture housing responses to credit-related shocks. Similarly, we do not control for 2008(4), as it is part of the crisis period whose dynamics we aim to forecast. Introducing such dummies could therefore remove relevant variation from our forecast.

4.5 The preferred model

Our preferred model is AR(6) without dummies (Table 4). Housing starts are highly persistent: the first lag has a coefficient of 1.256, and the AR coefficients sums to 0.887, while the significant negative third and sixth lags generate cyclical dynamics. All inverse roots lie inside the unit circle (Table 3); hence the process is stationary, but only just: $\mid \lambda_{\max} \mid = 0.916$, implying a half life of:

$$H_{\frac{1}{2}} = \frac{\log(0.5)}{\log(0.916)} = 8.00 \text{ Quarters}$$

And the unconditional mean;

$$\hat{u} = \frac{\delta}{1 - \sum_{j=1}^{6} \theta_j} = 7.293 \tag{8}$$

corresponds to about 1,471,000 housing starts, roughly 2.8 times the 2009(1) level. Since the dominant inverse roots form a complex pair, the forecasts return to the mean in an oscillatory manner, producing

the overshooting discussed in Section 5.0.

5 Forecasts

Figure 4 shows the forecasts from the $AR(6)$ model. The forecasts oscillate as they converge towards the unconditional mean, suggesting the roots of the characteristic polynomial are complex (**Table 3**). The model overestimates the number of housings starts in every period of the forecast horizon, implying a systematic overestimation of the speed of the recovery. Moreover, the realised values between 2010 and 2014 fall outside the forecast intervals, indicating that the $AR(6)$ model performs poorly in predicting the recovery after the financial crisis.

For comparison, we consider the AR(3) model (Figure 5), which captures the recovery better. The $AR(3)$ model also overestimates the pace of the recovery, where its forecasts are above the realized number of new privately owned housing units throughout the horizon. However, a larger share of the realised values falls within the forecast intervals. Nonetheless, the overestimation remains systematic.

However, this comparison relies on information unavailable at the time of the forecasting. In 2009, the $AR(6)$ would have remained the preferred specification, substantially overestimating the speed of the housing market recovery.

6 Discussion

6.1 Why did we end up with a slow recovery?

Three factors may help explain the slower-than-predicted recovery. First, credit conditions tightened substantially after the financial crisis. Stricter lending standards made it more difficult for households and developers to obtain loans, and fewer approved loans for new construction translated directly into fewer new privately owned housing units. Second, the housing market was in a state of disequilibrium with a large overhang of vacant homes. Many of these had been left empty through foreclosures on mortgages that could not be repaid, and this existing stock had to be absorbed before demand for new construction could recover. Finally, income growth remained weak in the years following the crisis, which, together with a weak labour market, depressed household formation and further dampened the demand for new housing units (Joint Center for Housing Studies, 2012; Paciorek, 2013; Board of Governors of the Federal Reserve System, 2012).

6.2 The models limitations

The factors discussed above highlight an important limitation, namely (A5). For the dynamic forecast to be informative, the relationship from the pre-2009 sample must remain valid during the forecast. However, the systematic overprediction of the recovery suggests that this assumption may not hold. Although $AR(6)$ is preferred based on the in-sample diagnostics, AR(3) performs better out of sample. This suggests that the richer dynamics captured by the AR(6) may describe historical housing cycles well but were less representative of the post-2009 recovery. Our comparison illustrates that good in-sample specification does not necessarily translate into superior out-of-sample forecasting performance when the underlying economic process changes

7 Conclusion

A univariate autoregression estimated on 1961(1)-2009(1) did not accurately forecast the US housing recovery. Our forecast with $AR(6)$ was rejected between 2010 and 2014, and the simpler $AR(3)$ which forecast better, but still overestimated the pace of recovery. This failure is systematic, suggesting that the historical relationship between housing and their long-run level was not fully representative of the post 2009 period.

8 Figures



Figure 1: $y_t = \log(HOUST_t)$

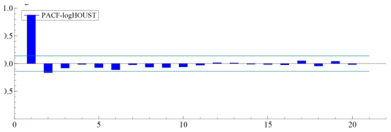


Figure 2: PACF

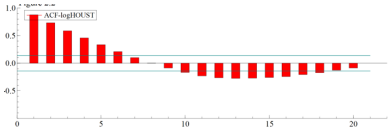


Figure 3: ACF

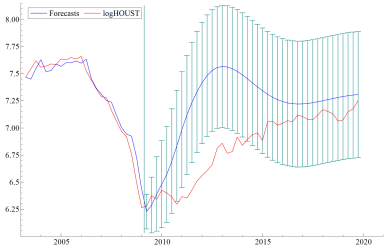


Figure 4: AR(6)

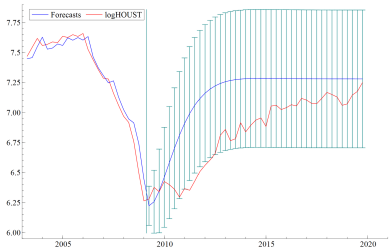


Figure 5: AR(3)

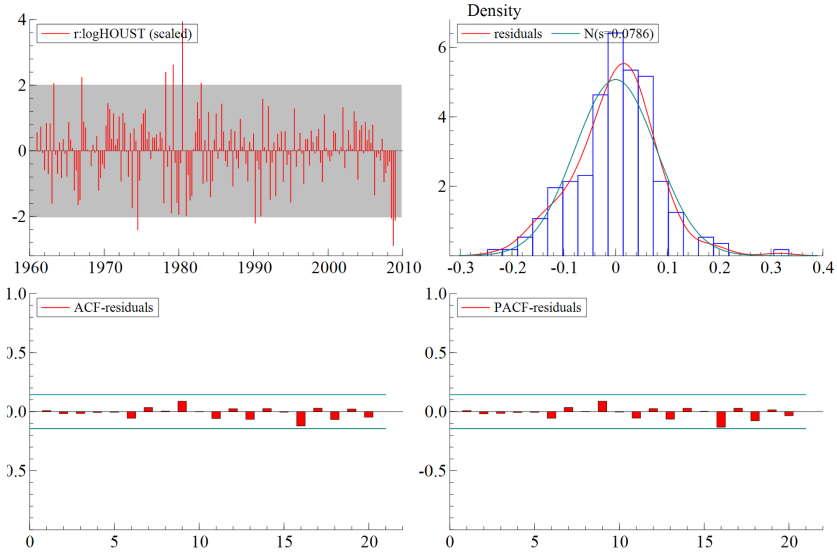


Figure 6: Residuals & Distribution

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